Lecturer: **Dr. Dmitriy Dunaev**

Final Project Selection Form

1. General Information

Project title: 3D Graphic Rendering in C++

Student

Full name: Nguyen Ba Phi

Neptun code: S3VYH3

2. Honor Code

By filling and submitting this form I certify that all of the following are true:

1) I fully understand the consequences of plagiarism and cheating.
2) I fully understand the meaning of plagiarism and recognize specifically that it includes copying of others' source code, paraphrasing, reusing old final projects, source code, and related materials.
3) I recognize that the minimum penalty for plagiarism and cheating with the final project is a Fail (1) in the course.
4) If I am unsure about whether something constitutes plagiarism, I will consult my instructor before I turn in the project assignment.
5) I will give the correct information on this form.

Please, answer questions completely. If approved this form becomes part of your Final Project Assignment.

All information provided will be kept confidential from other students.

3. General Requirements for Final Project

The Final Project requires the application of system and algorithm design, testing, and documentation skills. The minimal requirement includes the application of object-oriented paradigm, dynamic memory management (new/delete), exception handling (try/catch), and file management. The lab instructor can award extra point for additional C++ techniques, e.g.:

- Inheritance: base and derived classes
- Polymorphism: abstract classes and virtual functions.
- Multiple Inheritance: two or more base classes
- Operator overloading with local and global operations
- Generic data structures: class templates and function templates

You do not have to select the extra techniques now. Later, in project documentation, you are supposed to clearly explain what techniques from this list you might have implemented. This will be checked at personal defense.

4. Questions about the Project

1. Give a general description of the problem domain, provide theoretical and/or empirical backgrounds

The problem domain focuses on volumetric rendering and simulated light transport for 3D graphics. Specifically it explores how to simulate a physical volumetric display where discrete light projectors emit rays into a scattering medium to create 3D objects through additive light intersection. Theoretical backgrounds involve the physics of light absorption, exponential mist transparency falloff, and isotropic scattering. Simulating this efficiently requires mathematical models for 3D ray marching, discrete voxel grids, and energy conservation during secondary light bounces to prevent infinite feedback loops.

2. Describe your solution to the above problem (algorithms, methodology, reasoning):

The solution is an object-oriented C++ volumetric rendering engine. The core data structure is a dynamically allocated 3D grid of voxels containing color, transparency, and brightness values. For light transport I implemented a 3D Digital Differential Analyzer algorithm to march rays from virtual projectors through the voxel space calculating energy absorption based on mist transparency. For global illumination a breadth-first ping-pong buffer calculates secondary light scattering while strictly enforcing energy conservation. The system also supports parametric object generation using a 3D Bresenham line algorithm and object transformations using inverse rotation matrices to prevent backward mapping

3. Are you planning to use external solutions, e.g. class libraries, interfaces, engines? Explain below.

No external engines or third-party class libraries will be used. The entire engine including the ray marcher, 3D math, and ASCII renderer is built from scratch using the C++ Standard Library. The project relies strictly on standard libraries such as iostream, fstream for file management, vector for data containers, cmath for 3D mathematics, and random for probabilistic ray generation.

4. What IDE running on what OS will you be using for writing code?

The code is being written and compiled on macOS Apple Silicon. The development environm

5. Notes to the lab instructor

None.

Please note that only programs written in C++ with an object-oriented approach that meet the requirements can be accepted.

If you are unsure about your idea for the final project, seek guidance from your lab instructor before submitting this form.

Upload this form only at the Course Website; do not send it by e-mail or in any other way.

The submission deadline can be found on the Course website. No submissions will be accepted after the deadline.